

SARATH K BALAN

MERN Stack Developer

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PROFESSIONAL SUMMARY

Full-Stack MERN Developer with professional experience building production web applications using MongoDB, Express.js, React.js, and Node.js at TrackGenesis. Skilled in REST API design, authentication/authorization, reusable UI components, and AI-assisted feature integration. Comfortable working across the stack, from database schema and backend services to responsive React interfaces. Also experienced with React Native, NestJS, and blockchain (Solidity/Ethereum), with a growing background in the Microsoft stack (C#, ASP.NET Core, SQL Server).

TECHNICAL SKILLS

- Frontend: React.js, React Native, Redux/Context patterns, Responsive UI Design, HTML5, CSS3
- Backend: Node.js, Express.js, NestJS, REST API Design, Authentication & Authorization (JWT)
- Database: MongoDB, SQL Server
- Languages: JavaScript, C#, Python, Solidity
- AI/LLM: LLM Integration, AI APIs, Prompt Engineering, AI-Assisted Feature Development, Automation
- Tools & Practices: Git, GitHub, Docker, Postman, Swagger, VS Code, OOP, Clean Architecture, Agile
- Blockchain (additional): Ethereum, Solidity, Hardhat, Ethers.js, IPFS, Hyperledger Fabric

PROFESSIONAL EXPERIENCE

MERN Software Developer — TrackGenesis

2025 – Present

- Developed full-stack web applications using React.js, Node.js, Express.js, MongoDB, and REST APIs, covering both client-side interfaces and server-side services.
- Built reusable, component-based UI features in React.js and implemented backend APIs, authentication, authorization, and database-driven functionality.
- Worked with NestJS and React Native to extend features across web and mobile, and integrated AI-assisted functionality into dashboards, automation, and scheduling workflows.
- Collaborated across the stack on feature design, debugging, and performance improvements in an Agile team environment.

PROJECTS

Energy Trading Platform

MERN, AI

- Built an energy marketplace with consumer, prosumer, and generator roles using React, Node.js, and MongoDB.
- Developed role-based dashboards, marketplace and rate-management features, automated scheduling, and AI-generated insights.

Certificate Application System

React, Blockchain

- Built a decentralized certificate verification dApp with a React front end, using Solidity, Hardhat, Ethers.js, Ethereum, and IPFS, deployed to the Sepolia testnet with MetaMask integration.

Plant Selling & Inventory Management Platform

React, ASP.NET Core, SQL Server

- Developed the React front end for a full-stack e-commerce platform with customer, seller, and admin roles, consuming REST APIs for plant listings, cart, orders, and inventory.
- Implemented JWT-based authentication and role-based authorization across the application.

EDUCATION

Diploma in Blockchain — Kerala Blockchain Academy

2024 – 2025

B.Tech in Industrial Engineering — College of Engineering, Trivandrum

2017 – 2021 | CGPA: 7.09

CERTIFICATIONS

- Developer Essentials for Blockchain — Kerala Blockchain Academy
- AI Fundamentals and the Cloud (AWS); Generative AI: Elevate Your Software Development Career (IBM)
- Machine Learning with Python (IBM); AI for Research and Insights (Google); AI for Brainstorming and Planning (Google)